Statin therapy, muscle function and falls risk in community-dwelling older adults.

BACKGROUND: Statin therapy can cause myopathy, however it is unclear whether this exacerbates age-related muscle function declines. AIM: To describe differences between statin users and non-users in muscle mass, muscle function and falls risk in a group of community-dwelling older adults. DESIGN: A prospective, population-based cohort study with a mean follow-up of 2.6 years. METHODS: Total 774 older adults [48% female; mean (standard deviation) age = 62 (7) years] were examined at baseline and follow-up. Differences in percentage appendicular lean mass (%ALM), leg strength, leg muscle quality (LMQ; specific force) and falls risk were compared for statin users and non-users. RESULTS: There were 147 (19%) statin users at baseline and 179 (23%) at follow-up. Longitudinal analyses revealed statin use at baseline predicted increased falls risk scores over 2.6 years (0.14, 95% CI 0.01 to 0.27) and a trend towards increased %ALM (0.45%, 95% CI -0.01 to 0.92). Statin users at both time points demonstrated decreased leg strength (-5.02 kg, 95% CI -9.65 to -0.40) and LMQ (-0.30 kg/kg, 95% CI -0.59 to -0.01), and trended towards increased falls risk (0.13, 95% CI -0.01 to 0.26) compared to controls. Finally, statin users at both baseline and follow-up demonstrated decreased leg strength (-16.17 kg, 95% CI -30.19 to -2.15) and LMQ (-1.13 kg/kg, 95% CI -2.02 to -0.24) compared to those who had ceased statin use at follow-up. CONCLUSION: Statin use may exacerbate muscle performance declines and falls risk associated with aging without a concomitant decrease in muscle mass, and this effect may be reversible with cessation.